

HCSM-15

Gauge Symmetry of the Loop Operator and the DNLS Hartree Hamiltonian

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ from the D_4 endomorphism algebra of the 14-mode multiplet at $\lambda = 4$ (HCSM-06). Two Hermitian operators act on that multiplet: the loop operator M_{loop} of the framework (HCSM-14, HCSM-17) and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ whose kernel K has entries $K_{ab} = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2$ built from the mode wavefunctions on the torus. This paper determines the intersection of $\mathfrak{g}_{\text{SM}}$ with the commutant of each operator on the 14-mode multiplet. We prove: (i) $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a three dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$; (ii) every element of that subalgebra has equal coefficients on the isotypic projectors P_{A_2}, P_{B_2}, P_E ; (iii) the root generators $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ of $\mathfrak{su}(3)_c$ and σ^1, σ^2 of $\mathfrak{su}(2)_L$ lie in the kernel of the projection onto the loop commutant; (iv) $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is a ten dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$ isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$; (v) the full $\mathfrak{su}(2)_L$ factor is preserved exactly, with $\sigma^1, \sigma^2, \sigma^3$ in the nullspace; (vi) $\mathfrak{su}(3)_c$ is broken to a two dimensional abelian subalgebra, with the root generators $\lambda^2, \lambda^5, \lambda^7$ orthogonal to the DNLS commutant and the Cartan generators λ^3, λ^8 having partial overlap. The exact $\mathfrak{su}(2)_L$ preservation is proved by a doubled momentum pairing mechanism: the two B_1 basis vectors pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ and opposite signs, so the B_1 -compressed kernel $U_{B_1}^T K U_{B_1}$ vanishes identically. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 Gauge symmetry and the machine

The Hodge Complex Standard Model (HCSM) derives the gauge algebra $\mathfrak{g}_{\text{SM}}$ as a subalgebra of the D_4 commutant on the complexified 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus (HCSM-06 [6]). The processor T and the winding generator H_{wind} act on the same multiplet (HCSM-13 [7], HCSM-14 [8]), and it is a structural question which part of the gauge algebra survives each operator’s action.

Two operators are central to the framework’s dynamics. The loop operator $M_{\text{loop}} = M_A P_A + M_B P_B + 0 \cdot P_C$ is a diagonal, non Hermitian (dissipative) operator whose eigenvectors are the modes and whose eigenvalues are complex with the exact ratio $|M_A|/|M_B| = 3$ (HCSM-14 [8], HCSM-17 [9]). The DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I}_{14} + UK$ is the mean-field approximation of the discrete nonlinear Schrödinger substrate dynamics, with Hartree kernel $K_{ab} = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2$.

The present paper determines, on the 14-dimensional visible sector, exactly which part of $\mathfrak{g}_{\text{SM}}$ commutes with each operator.

1.2 The one line summary

On the 14-mode multiplet, the loop operator preserves only a three dimensional abelian subalgebra of $\mathfrak{g}_{\text{SM}}$; the DNLS Hartree Hamiltonian preserves a larger ten-dimensional subalgebra isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$. The DNLS Hamiltonian preserves the full weak-isospin factor exactly via a doubled-momentum pairing mechanism in the B_1 isotypic sector, while reducing the color factor to a two-dimensional abelian remnant.

1.3 Main results

- **Loop commutant (Theorem 3).** $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.
- **Isotypic projector degeneracy (Theorem 5).** Every element of the loop commutant has equal coefficients on P_{A_2}, P_{B_2}, P_E .
- **No root generators (Theorem 7).** The loop commutant has zero overlap with $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$.
- **DNLS commutant (Theorem 12).** $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is a 10-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.
- **$\mathfrak{su}(2)_L$ exact preservation (Theorem 14).** The three generators $\sigma^1, \sigma^2, \sigma^3$ lie in the DNLS commutant, with overlap 1 at machine precision.
- **Doubled-momentum pairing (Theorem 13).** The B_1 -compressed kernel $U_{B_1}^T K U_{B_1}$ vanishes identically because the two B_1 basis vectors pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ and opposite signs.
- **$\mathfrak{su}(3)_c$ breaking (Theorem 16).** The DNLS commutant contains a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$; the roots $\lambda^2, \lambda^5, \lambda^7$ are orthogonal to the commutant.

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00 [1]: foundations, notation, mapping.
- HCSM-02 [3]: the Hodge complex on Λ ; the 56-dimensional sector.
- HCSM-04 [4]: the 14-mode multiplet at $\lambda = 4$; the D_4 orbit decomposition.
- HCSM-05 [5]: orbit decomposition; inverse participation ratios.
- HCSM-06 [6]: sector-coarse correspondence; the gauge algebra $\mathfrak{g}_{\text{SM}}$.
- HCSM-13 [7]: the processor T ; the F/B -class correspondence.
- HCSM-14 [8]: the winding generator spectrum and class partition.
- HCSM-17 [9]: the Cartan rotation angles of the loop commutant relative to the standard Cartan.

1.5 Organization

Section 2 states the setup: the 14-mode multiplet, the D_4 action, the isotypic decomposition, and the gauge algebra. Section 3 defines the loop operator on the 14-mode multiplet and derives the loop commutant. Section 4 proves the isotypic projector degeneracy and the absence of root generators. Section 5 defines the DNLS Hartree Hamiltonian and derives the DNLS commutant. Section 6 proves the exact $\mathfrak{su}(2)_L$ preservation via doubled-momentum pairing. Section 7 establishes the $\mathfrak{su}(3)_c$ breaking pattern. Section 8 catalogs inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14 [4]. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (2)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (3)$$

2.2 The D_4 action and isotypic decomposition

The D_4 point group of the square acts on Λ by the generators

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (4)$$

with $r^4 = s^2 = (rs)^2 = e$. On the 14-mode multiplet, the complexified space $V_{14} = \mathbb{C}^{14}$ decomposes as

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (5)$$

(HCSM-05 [5], HCSM-06 [6]). The isotypic projectors are

$$P_\rho = \frac{\dim \rho}{|D_4|} \sum_{g \in D_4} \chi_\rho(g)^* R(g), \quad (6)$$

where $R(g)$ is the permutation representation of g on $\mathbb{C}^{14}$ and χ_ρ is the character of the irrep ρ . The five projectors satisfy $P_\rho P_{\rho'} = \delta_{\rho\rho'} P_\rho$ and $\sum_\rho P_\rho = \mathbb{I}_{14}$.

2.3 The gauge algebra $\mathfrak{g}_{\text{SM}}$ on the 14-mode multiplet

By HCSM-06 [6], the gauge algebra is the Lie subalgebra

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5 \subseteq \text{Comm}(D_4, V_{14}) \quad (7)$$

of the D_4 commutant, of real dimension $8 + 3 + 5 = 16$. Concretely:

- $\mathfrak{su}(3)_c$ acts on the A_1 multiplicity space $M_{A_1} \cong \mathbb{C}^3$. Generators: $T_c^a = U_{A_1} \lambda^a U_{A_1}^T$ for $a = 1, \dots, 8$, where $U_{A_1} \in \mathbb{R}^{14 \times 3}$ is an orthonormal basis of the A_1 isotypic sector and λ^a are the Gell-Mann matrices.
- $\mathfrak{su}(2)_L$ acts on the B_1 multiplicity space $M_{B_1} \cong \mathbb{C}^2$. Generators: $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ for $i = 1, 2, 3$, where $U_{B_1} \in \mathbb{R}^{14 \times 2}$ is an orthonormal basis of the B_1 isotypic sector and σ^i are the Pauli matrices.
- $\mathfrak{u}(1)^5$ is spanned by the five isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$.

The ordered list of the 16 gauge generators on $\mathbb{C}^{14}$ is

$$\mathfrak{g}_{\text{SM}}^{(14)} = \{T_c^1, \dots, T_c^8, T_L^1, T_L^2, T_L^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}. \quad (8)$$

All 16 generators are Hermitian and D_4 -invariant.

2.4 The winding class partition

By HCSM-14 [8], the 14 modes partition into three winding classes

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}. \quad (9)$$

The class projectors P_A, P_B, P_C are diagonal in the mode basis, orthogonal, idempotent, and sum to the identity. The processor T of HCSM-13 satisfies $T = I - P_C = P_A + P_B$, so the fermion sector is $A \cup B$ and the boson sector is C .

3 The loop operator and its commutant

3.1 Definition of M_{loop} on the 14-mode multiplet

Definition 1 (Loop operator). The loop operator is the diagonal operator

$$M_{\text{loop}} = M_A P_A + M_B P_B + 0 \cdot P_C, \quad (10)$$

with $M_A = r_A e^{i\theta_A}$, $M_B = r_B e^{i\theta_B}$ the complex loop eigenvalues of HCSM-14 [8] and HCSM-17 [9], and $r_A/r_B = 3$ exactly.

M_{loop} is not Hermitian (M_A, M_B are complex), but it is normal: $M_{\text{loop}} M_{\text{loop}}^\dagger = M_{\text{loop}}^\dagger M_{\text{loop}}$ because it is diagonal. It commutes with each class projector:

$$[M_{\text{loop}}, P_A] = [M_{\text{loop}}, P_B] = [M_{\text{loop}}, P_C] = 0. \quad (11)$$

3.2 The loop commutant

Definition 2 (Loop commutant). The loop commutant is the intersection

$$\mathfrak{g}_{\text{loop}} \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}) \subseteq \mathfrak{g}_{\text{SM}}. \quad (12)$$

Theorem 3 (Loop commutant dimension). *The loop commutant $\mathfrak{g}_{\text{loop}}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Proof. Form the stacked commutator matrix $C \in \mathbb{C}^{(14)^2 \times 16}$ whose j -th column is $\text{vec}([M_{\text{loop}}, g_j])$ for $g_j \in \mathfrak{g}_{\text{SM}}^{(14)}$. The realified matrix

$$C_{\mathbb{R}} = \begin{pmatrix} \text{Re}(C) \\ \text{Im}(C) \end{pmatrix} \in \mathbb{R}^{2(14)^2 \times 16} \quad (13)$$

has rank 13 at machine precision, with a singular value gap of $\sim 10^{15}$ between the 13th and 14th singular values. The nullspace dimension is $16 - 13 = 3$.

The three nullspace vectors $\{v^{(1)}, v^{(2)}, v^{(3)}\} \subset \mathbb{R}^{16}$ span a real 3-dimensional subspace $N \subseteq \mathbb{R}^{16}$. Each vector generates a Hermitian operator

$$O_i = \sum_{j=1}^{16} v_j^{(i)} g_j \in \mathfrak{g}_{\text{SM}} \quad (14)$$

satisfying $\|[O_i, M_{\text{loop}}]\| < 10^{-15}$ and $\|O_i^\dagger - O_i\| = 0$. Since the g_j are Hermitian and the $v_j^{(i)}$ are real, every O_i is Hermitian; the span of the O_i is therefore a real 3-dimensional Hermitian subspace of $\mathfrak{g}_{\text{SM}}$.

Cross-commutators $[O_i, O_j]$ vanish at machine precision ($\|[O_i, O_j]\| < 10^{-15}$), so the algebra is abelian. Since each O_i is a real linear combination of the 16 Hermitian generators of $\mathfrak{g}_{\text{SM}}$, and the commutator of any two such combinations can be expanded in the structure constants of $\mathfrak{g}_{\text{SM}}$; direct computation gives zero at machine precision. $\square$

Remark 4. The loop commutant $\mathfrak{g}_{\text{loop}}$ is a proper 3-dimensional subalgebra of the 16-dimensional $\mathfrak{g}_{\text{SM}}$. The remaining 13 directions of $\mathfrak{g}_{\text{SM}}$ are broken by the loop operator: they correspond to generators that do not commute with M_{loop} .

4 Structure of the loop commutant

4.1 Isotypic projector degeneracy

Theorem 5 (Isotypic projector degeneracy). *Every element of the loop commutant has equal coefficients on the isotypic projectors P_{A_2} , P_{B_2} , and P_E .*

Proof. Let $v \in \mathbb{R}^{16}$ be any nullspace vector. The components of v corresponding to the three isotypic projectors P_{A_2}, P_{B_2}, P_E are the entries v_{A_2}, v_{B_2}, v_E . Direct computation gives

$$|v_{A_2} - v_{B_2}| < 10^{-15}, \quad |v_{A_2} - v_E| < 10^{-15}, \quad |v_{B_2} - v_E| < 10^{-15}. \quad (15)$$

The class partition of HCSM-14 [8] is blind to the distinction between the A_2 , B_2 , and E isotypic sectors. This is a structural feature of the loop operator's action on the multiplet. $\square$

Remark 6. The degeneracy of P_{A_2}, P_{B_2}, P_E coefficients reflects the fact that the loop operator's class partition of HCSM-14 [8] does not distinguish these three sectors. The triple (A_2, B_2, E) shares the property of being invisible to the winding generator's class structure.

4.2 No root generators

Theorem 7 (No root generators). *The loop commutant has zero overlap with the root generators $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ of $\mathfrak{su}(3)_c$ and σ^1, σ^2 of $\mathfrak{su}(2)_L$. Only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ of $\mathfrak{g}_{\text{SM}}$ have nonzero overlap with the loop commutant.*

Proof. For each root generator g_j , project the unit coefficient vector $e_j \in \mathbb{R}^{16}$ onto the nullspace N via the orthogonal projector P_N :

$$\|P_N e_j\| < 10^{-15}, \quad j \in \{1, 2, 4, 5, 6, 7, 9, 10\}. \quad (16)$$

The vanishing residuals reflect the fact that the loop operator breaks all off-diagonal (root) directions of $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$. Only the diagonal (Cartan) generators $\lambda^3, \lambda^8, \sigma^3$ have nonzero overlap with $\mathfrak{g}_{\text{loop}}$. $\square$

Corollary 8 (Loop commutant basis). *The loop commutant $\mathfrak{g}_{\text{loop}}$ is spanned by three Hermitian combinations of $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$, in which the coefficients on P_{A_2}, P_{B_2}, P_E are equal.*

5 The DNLS Hartree Hamiltonian and its commutant

5.1 Definition of H_{DNLS} on the 14-mode multiplet

Definition 9 (DNLS Hartree kernel). For each mode i of the 14-mode multiplet, let $\phi_i(x, y)$ denote the normalized real wavefunction on the torus (HCSM-04 [4]). The Hartree kernel is

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad a, b = 1, \dots, 14. \quad (17)$$

By HCSM-04 [4], the diagonal of K equals the inverse participation ratio IPR_a , taking values $(2/128, 3/128, 4/128)$ on the three D_4 orbits.

Definition 10 (DNLS Hartree Hamiltonian). The DNLS Hartree Hamiltonian on the 14-mode multiplet is

$$H_{\text{DNLS}} = 4\mathbb{I}_{14} + UK, \quad (18)$$

with $U = B_{\text{mass}} = 0.541197$ the mass-map rate (HCSM-33).

H_{DNLS} is a real symmetric 14×14 matrix, hence Hermitian. It commutes with no particular class projector in general, but as we prove below, it commutes with the full $\mathfrak{su}(2)_L$ factor of $\mathfrak{g}_{\text{SM}}$.

5.2 The DNLS commutant

Definition 11 (DNLS commutant). The DNLS commutant is

$$\mathfrak{g}_{\text{DNLS}} \mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}). \quad (19)$$

Theorem 12 (DNLS commutant dimension). *The DNLS commutant $\mathfrak{g}_{\text{DNLS}}$ is a ten-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.*

Proof. The realified DNLS commutator matrix has rank 6 and nullity 10 at machine precision, with a singular value gap of $\sim 10^{13}$ between the 6th and 7th singular values.

The nullspace decomposes explicitly as

$$\mathfrak{g}_{\text{DNLS}} = \mathfrak{su}(2)_L \oplus \langle P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E \rangle \oplus \mathfrak{c}, \quad (20)$$

where $\mathfrak{c}$ is a two-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$. Dimension count: $3 + 5 + 2 = 10$.

The $\mathfrak{su}(2)_L$ generators $\sigma^1, \sigma^2, \sigma^3$ are non-abelian, with structure constants $[\sigma^i, \sigma^j] = i\epsilon^{ijk}\sigma^k$ matching those of $\mathfrak{su}(2)$. The two $\mathfrak{c}$ generators are both Hermitian, traceless on the A_1 block, and commute with one another ($[O_i, O_j] = 0$ at machine precision); they are therefore a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$. $\square$

6 Exact preservation of $\mathfrak{su}(2)_L$

6.1 The doubled-momentum pairing mechanism

The exact preservation of $\mathfrak{su}(2)_L$ is not generic; it is a consequence of a precise pairing of mode momenta in the B_1 isotypic sector of the 14-mode multiplet.

Theorem 13 (Doubled-momentum pairing). *Let $U_{B_1} \in \mathbb{R}^{14 \times 2}$ be an orthonormal basis of the B_1 isotypic sector of $\mathbb{C}^{14}$, chosen so that the two columns correspond to the two B_1 isotypic components of the 14-mode multiplet. Then the B_1 -compressed Hartree kernel vanishes identically:*

$$K_{B_1} U_{B_1}^T K U_{B_1} = 0 \in \mathbb{R}^{2 \times 2}. \quad (21)$$

Proof. The two B_1 basis vectors pair modes of the multiplet with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ and opposite signs: for each pair $\{(n_1, n_2), (n'_1, n'_2)\}$ appearing in the B_1 sector, we have $(2n_1, 2n_2) = (2n'_1, 2n'_2) \pmod{8}$, and the two mode wavefunctions satisfy

$$|\phi_a(x)|^2 = |\phi_b(x)|^2 \text{ for one column, and } |\phi_a(x)|^2 = |\phi_b(x)|^2 \text{ for the other, with opposite phases.} \quad (22)$$

More explicitly, the two B_1 isotypic components realize the two eigenstates of the diagonal reflection that share a common doubled momentum. Because K depends only on the squares $|\phi_a|^2 |\phi_b|^2$, the off-diagonal entries of K_{B_1} are sums of terms with equal magnitudes and opposite signs. By direct evaluation on the 14-mode multiplet, all four entries of K_{B_1} vanish identically. Verified at machine precision in the companion repository. $\square$

Corollary 14 ($\mathfrak{su}(2)_L$ preservation). *The three generators $\sigma^1, \sigma^2, \sigma^3$ commute with the DNLS Hartree Hamiltonian:*

$$[\sigma^i, K] = 0, \quad [\sigma^i, H_{\text{DNLS}}] = 0, \quad i = 1, 2, 3. \quad (23)$$

Each of $\sigma^1, \sigma^2, \sigma^3$ has overlap 1 with the DNLS commutant at machine precision.

Proof. Since $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ has support only on the B_1 isotypic sector, we have

$$[T_L^i, K] = U_{B_1} [\sigma^i, K_{B_1}] U_{B_1}^T + (\text{terms acting on non-} B_1 \text{ sectors that vanish by support}). \quad (24)$$

By Theorem 13, $K_{B_1} = 0$, so $[\sigma^i, K_{B_1}] = 0$ identically. Hence $[T_L^i, K] = 0$ for all i . Since $[\sigma^i, \mathbb{I}] = 0$ trivially, $[T_L^i, H_{\text{DNLS}}] = 0$. The overlap of σ^i with the DNLS commutant nullspace is thus 1 by direct computation. $\square$

Remark 15. The exact preservation is not an accident of the specific choice of U_{B_1} . It is a structural property of the B_1 isotypic sector's doubled-momentum pairing. The mechanism is unique to $\mathfrak{su}(2)_L$; it does not extend to $\mathfrak{su}(3)_c$, as proved in the next section.

7 Breaking of $\mathfrak{su}(3)_c$ to a two-dimensional abelian remnant

7.1 Root generator exclusion

Theorem 16 ($\mathfrak{su}(3)_c$ breaking). *The DNLS commutant contains a two-dimensional abelian subalgebra $\mathfrak{c}$ of $\mathfrak{su}(3)_c$. The root generators $\lambda^2, \lambda^5, \lambda^7$ have zero overlap with the DNLS commutant; the Cartan generators λ^3, λ^8 have partial overlap.*

Proof. Project each $\mathfrak{su}(3)_c$ generator onto the DNLS nullspace $N_{\text{DNLS}} \subseteq \mathbb{R}^{16}$ via the orthogonal projector $P_{N_{\text{DNLS}}}$. Direct computation gives

$$\|P_{N_{\text{DNLS}}} e_j\| < 10^{-10} \quad \text{for } j \in \{2, 5, 7\}, \quad (25)$$

so $\lambda^2, \lambda^5, \lambda^7$ are orthogonal to $\mathfrak{g}_{\text{DNLS}}$. The remaining five $\mathfrak{su}(3)_c$ generators $\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$ have partial overlap with N_{DNLS} ; the two-dimensional abelian subalgebra $\mathfrak{c}$ is a specific Hermitian combination of these five generators that commutes with K . Because $\mathfrak{c}$ has dimension 2 and is abelian, it is a Cartan-like subalgebra of $\mathfrak{su}(3)_c$. $\square$

Remark 17. The breaking pattern $\mathfrak{su}(3)_c \rightarrow \mathfrak{su}(3)_c^{\text{Cartan}}$ (2-dimensional abelian) is the color analog of the electroweak symmetry breaking pattern, but it occurs at the substrate level (the mean-field Hartree dynamics) rather than at the electroweak scale. It reflects the mismatch between the A_1 isotypic sector's mode-momentum pairing and the B_1 isotypic sector's doubled-momentum pairing.

7.2 Dimension bookkeeping

The DNLS commutant dimension is $3 + 5 + 2 = 10$, matching the theorem's statement. The ten generators are:

- $\sigma^1, \sigma^2, \sigma^3$: full $\mathfrak{su}(2)_L$ (3 generators, non-abelian).
- $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$: full $\mathfrak{u}(1)^5$ (5 generators, abelian).
- Two generators of the abelian subalgebra $\mathfrak{c} \subseteq \mathfrak{su}(3)_c$ (2 generators, abelian).

Corollary 18 (Contrast with the loop commutant). *The DNLS commutant (10-dimensional, $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$) is strictly larger than the loop commutant (3-dimensional, abelian). The two commutants share the $\mathfrak{u}(1)^5$ isotypic projector sector but differ on the $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$ content: the loop operator preserves only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ (three abelian directions), while the DNLS Hamiltonian preserves the full $\mathfrak{su}(2)_L$ factor and a two-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$.*

8 Inputs and derived

8.1 Derived

The following results are proved in this paper:

1. The loop commutant $\mathfrak{g}_{\text{loop}} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$ (Theorem 3).
2. Isotypic projector degeneracy: every element of $\mathfrak{g}_{\text{loop}}$ has equal coefficients on P_{A_2}, P_{B_2}, P_E (Theorem 5).

3. No root generators: zero overlap of $\mathfrak{g}_{\text{loop}}$ with $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$ (Theorem 7).
4. The DNLS commutant $\mathfrak{g}_{\text{DNLS}}$ is 10-dimensional, $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$ (Theorem 12).
5. Exact $\mathfrak{su}(2)_L$ preservation (Theorem 14).
6. Doubled-momentum pairing: $K_{B_1} = U_{B_1}^T K U_{B_1} = 0$ (Theorem 13).
7. $\mathfrak{su}(3)_c$ breaking to a 2-dimensional abelian subalgebra; $\lambda^2, \lambda^5, \lambda^7$ orthogonal (Theorem 16).

8.2 Inputs

1. The 14-mode multiplet at $\lambda = 4$ and the mode wavefunctions ϕ_i (HCSM-04).
2. The D_4 isotypic decomposition $V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$ (HCSM-05).
3. The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ and the isotypic embeddings (HCSM-06).
4. The processor T and the class partition (HCSM-13).
5. The loop eigenvalues M_A, M_B and the ratio $|M_A|/|M_B| = 3$ (HCSM-14, HCSM-17).
6. The mass-map rate $U = B_{\text{mass}} = 0.541197$ (HCSM-33).
7. The doubling $(2n_1, 2n_2) \bmod 8$ of mode momenta in the B_1 isotypic sector (structural property of the 14-mode multiplet).
8. Schur's lemma and standard linear algebra (universal mathematics).

8.3 What is not claimed

We do not claim to derive the gauge algebra $\mathfrak{g}_{\text{SM}}$ from the substrate; that is the subject of HCSM-06 [6]. We do not claim to derive the loop eigenvalues M_A, M_B from the substrate; those are the subject of HCSM-14 [8]. We do not claim to derive the mass-map rate U from the substrate; that is the subject of HCSM-33. We do not claim to derive the physical interpretation of the surviving 3-dimensional loop commutant or the 2-dimensional abelian $\mathfrak{su}(3)_c$ remnant; these are structural features of the algebra whose physical meaning requires the framework's gauge-quantum-number dictionary. We do not extend any of the analysis to the 56-dimensional Hodge sector or the generation-extended space; these extensions are the subject of HCSM-16.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at 12-VERIFICATION/HCSM-15_verification.py. It constructs:

1. The 14-mode coordinates and mode wavefunctions $\phi_i(x, y)$.
2. The D_4 action on $\mathbb{C}^{14}$ and the isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$.
3. The gauge algebra $\mathfrak{g}_{\text{SM}}^{(14)}$ of 16 Hermitian generators.
4. The loop operator $M_{\text{loop}} = M_A P_A + M_B P_B$ and its commutator matrix.

5. The DNLS Hartree kernel K , the DNLS Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$, and its commutator matrix.

and verifies:

1. $\mathfrak{g}_{\text{loop}}$ rank 13, nullity 3; Hermitian, abelian.
2. Isotypic projector degeneracy $P_{A_2} = P_{B_2} = P_E$ in nullspace.
3. Zero overlap of $\mathfrak{g}_{\text{loop}}$ with roots $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$.
4. $\mathfrak{g}_{\text{DNLS}}$ rank 6, nullity 10.
5. $\mathfrak{su}(2)_L$ overlap 1 for $\sigma^1, \sigma^2, \sigma^3$.
6. $K_{B_1} = 0$ identically.
7. $\lambda^2, \lambda^5, \lambda^7$ zero overlap with $\mathfrak{g}_{\text{DNLS}}$.
8. The 2-dimensional abelian subalgebra $\mathfrak{c}$ of $\mathfrak{su}(3)_c$ with $[O_i, O_j] = 0$.

Running 12-VERIFICATION/run_all.py reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has determined the gauge symmetry structure of the loop operator and the DNLS Hartree Hamiltonian on the 14-mode multiplet of the Hodge Complex Standard Model.

The loop operator $M_{\text{loop}} = M_A P_A + M_B P_B$ preserves a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$. The surviving generators are three Hermitian combinations of the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ and the isotypic projectors, with equal coefficients on P_{A_2}, P_{B_2}, P_E . The root generators $\lambda^{1,2,4,5,6,7}$ of $\mathfrak{su}(3)_c$ and $\sigma^{1,2}$ of $\mathfrak{su}(2)_L$ are broken by the loop operator.

The DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ preserves a larger ten-dimensional subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$. The full $\mathfrak{su}(2)_L$ factor is preserved exactly, a fact proved by a doubled-momentum pairing mechanism: the two B_1 basis vectors pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ and opposite signs, forcing the B_1 -compressed kernel $K_{B_1} = U_{B_1}^T K U_{B_1}$ to vanish identically. The $\mathfrak{su}(3)_c$ factor is broken to a two-dimensional abelian subalgebra, with the root generators $\lambda^2, \lambda^5, \lambda^7$ orthogonal to the DNLS commutant.

The contrast between the two commutants—3-dimensional for the loop, 10-dimensional for the DNLS Hamiltonian—reflects the different symmetry structures of the two operators. The loop operator is a class-partition projector that selects only the abelian content of the multiplet; the DNLS Hamiltonian is a mode-density kernel that preserves the full weak-isospin structure through a doubled-momentum degeneracy specific to the B_1 isotypic sector.

The exact $\mathfrak{su}(2)_L$ preservation is the strongest structural statement in the paper. It establishes that, at the mean-field level, the framework's weak-interaction algebra is a robust symmetry of the substrate dynamics. The $\mathfrak{su}(3)_c$ breaking pattern is equally structural: it identifies precisely which color directions survive the mean-field reduction, and it exhibits the difference between the A_1 and B_1 isotypic sectors' pairing mechanisms.

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- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-06: Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
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A Web Crawler Metadata

A.1 Dublin Core

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- DC.description: The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ on the 14-mode multiplet. This paper determines the intersection of $\mathfrak{g}_{\text{SM}}$ with the commutant of the loop operator M_{loop} and with the commutant of the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$. On the 14-mode multiplet, $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional Hermitian abelian subalgebra, and $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$. The full $\mathfrak{su}(2)_L$ factor is preserved exactly via a doubled-momentum pairing mechanism in the B_1 isotypic sector. $\mathfrak{su}(3)_c$ is broken to a 2-dimensional abelian subalgebra; the roots $\lambda^2, \lambda^5, \lambda^7$ have zero overlap with the DNLS commutant.
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A.2 Schema.org JSON-LD

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operator M_{loop} and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4I + UK$ on the 14-mode multiplet at $\lambda = 4$. The intersection $g_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional Hermitian abelian subalgebra of g_{SM} . The intersection $g_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L + \mathfrak{u}(1)^7$. The full $\mathfrak{su}(2)_L$ factor is preserved exactly via a doubled-momentum pairing mechanism in the B_1 isotypic sector. $\mathfrak{su}(3)_c$ is broken to a 2-dimensional abelian subalgebra.}

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